



Shri Guru Gobind Singh Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Electrical Engineering
In-Semester Examination / Evaluation I
Academic Year: 2025-26 (semester I)

Class: Final Year B.Tech.

Course: Electrical Machine Design

Course Code: PCC-EE403

Date: 20.08.2025

Time: 15:00 to 15:45

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.	Attempt all questions.	Marks	CO	BT
Q.1	List out major differences between distribution and power transformer.	5	CO1	BT1
Q.2	Derive an expression for kVA output of single-phase transformer.	5	CO1	BT2
Q.3	A 3-Phase core type transformer of 400 kVA, 6600/440 V, 50 Hz, has 15 V emf per turn, 1.5 Wb/m ² maximum flux density, 2 A/mm ² current density, 0.2 window space factor, 0.9 stacking factor, use a 3 stepped core. For a three stepped core width of largest stamping = 0.9 d, net iron area = 0.6 d ² . where, d is the diameter of circumscribing circle. If transformer overall height is equal to overall width, then calculate approximate overall dimensions.	6	CO1	BT3
Q.4	a Explain the path of heat flow in a transformer with a neat diagram.	2	CO2	BT2
	b Find the specific heat dissipation due to convection of transformer oil if there is 15°C temperature difference of the surface relative to oil and 0.75 m height of dissipating surface. Also compare it with the specific heat dissipation due to radiation of air.	2	CO2	BT3

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Explain the constructional details and design aspects of various types of transformers and calculate parameters such as EMF per turn, core dimensions, and insulation. (Understand, Apply)
CO 2	Evaluate transformer performance using equivalent circuit modeling, leakage reactance, and temperature rise analysis, and design cooling systems like tanks and radiators. (Evaluate, Design)
CO 3	Describe the construction and design process of three-phase induction motors including stator and rotor components, and calculate parameters like turns per phase, air gap length, and rotor currents.
CO 4	Analyze the performance of three-phase induction motors using circle diagram and derive key parameters such as maximum torque, power factor, and dispersion coefficient. (Analyze, Evaluate)
CO 5	Explain the construction and design methodology of synchronous machines and estimate main dimensions using electric/magnetic loadings and short-circuit ratio. (Understand, Apply)
CO 6	Apply computer-aided design techniques and optimization approaches to develop improved designs of transformers and induction motors. (Apply, Design)



Class: Third Year B.Tech.

Course: Signals and Systems

Course Code: PCC-EE305

Date: 21.08.2025

Time: 15:00 to 15:45

Max. Marks: 20

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Q. No.	Question Statement	Marks	CO	BT
Q.1	Represent the following standard signals both mathematically and graphically: - Unit Step Signal - Unit Ramp Signal - Exponentially Decaying Signal	6	CO1	BT2
Q.2	A discrete-time signal $x(n]$ is shown below, sketch and label carefully each of the following signals: - Time Shifting: $x(n-2)$ - Amplitude Scaling: $2x(n)$	4	CO1	BT3 and BT6
Q.3	- Determine if $x(t) = \cos(t)$ is even signal or odd signal. - What is the condition for an LTI system to be stable?	2	CO1	BT2
Q.4	Given two discrete-time signals: $x(n) = [2, 1, 3]$ and $h(n) = [1, 0, -1]$. Compute the linear convolution $y(n) = x(n) * h(n)$ using tabular method.	4	CO2	BT3
Q.5	- Write the mathematical expression for continuous-time and for discrete-time convolution. - If $x(t)*h(t) = h(t)*x(t)$, what property of convolution does this represent? - If the input signal, $x(n)$ has M samples and the impulse response, $h(n)$ has N samples, how many samples will the output signal, $y(n)$ have after convolution?	4	CO2	BT1 and BT2

== ALL THE BEST ==



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Department of Electrical Engineering
In-Semester Examination / Evaluation I
Academic Year: 2025-26 (semester I)

Class: TY B.Tech.

Course: Electrical Machine Design

Course Code: PEC-EE305

Date: 20.08.2025

Time: 15:00 to 15:45

Max. Marks: 20

Note:

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- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.	Attempt all questions.	Marks	CO	BT
Q.1	List out major differences between distribution and power transformer.	5	CO1	BT1
Q.2	Derive an expression for kVA output of single-phase transformer.	5	CO1	BT2
Q.3	A 3-Phase core type transformer of 400 kVA, 6600/440 V, 50 Hz, has 15 V emf per turn, 1.5 Wb/m ² maximum flux density, 2 A/mm ² current density, 0.2 window space factor, 0.9 stacking factor, use a 3 stepped core. For a three stepped core width of largest stamping = 0.9 d, net iron area = 0.6 d ² . where, d is the diameter of circumscribing circle. If transformer overall height is equal to overall width, then calculate approximate overall dimensions.	6	CO1	BT3
Q.4	a Explain the path of heat flow in a transformer with a neat diagram.	2	CO2	BT2
	b Find the specific heat dissipation due to convection of transformer oil if there is 15°C temperature difference of the surface relative to oil and 0.75 m height of dissipating surface. Also compare it with the specific heat dissipation due to radiation of air.	2	CO2	BT3

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Bloom's Taxonomy (BT)

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Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Explain the constructional details and design aspects of various types of transformers and calculate parameters such as EMF per turn, core dimensions, and insulation. (Understand, Apply)
CO 2	Evaluate transformer performance using equivalent circuit modeling, leakage reactance, and temperature rise analysis, and design cooling systems like tanks and radiators. (Evaluate, Design)
CO 3	Describe the construction and design process of three-phase induction motors including stator and rotor components, and calculate parameters like turns per phase, air gap length, and rotor currents.
CO 4	Analyze the performance of three-phase induction motors using circle diagram and derive key parameters such as maximum torque, power factor, and dispersion coefficient. (Analyze, Evaluate)
CO 5	Explain the construction and design methodology of synchronous machines and estimate main dimensions using electric/magnetic loadings and short-circuit ratio. (Understand, Apply)
CO 6	Apply computer-aided design techniques and optimization approaches to develop improved designs of transformers and induction motors. (Apply, Design)



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Electrical Engineering
In-Semester Examination-I
Academic Year: 2025-26 (semester I)

Class: Third Year

Course: Feedback Control System

Course Code: PCC-EE302/303

Date: 19.08.2025

Time: 15:00 to 15:45 pm

Max. Marks: 20

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Q.N.		Marks	CO	BT
Q.1	Attempt any two questions.			
a	Differentiate an Open Loop and Closed Loop Control System	05	CO1	BT1
b	Explain in short SISO and MIMO.	05	CO1	BT2
c	Define the following terms : I) Time Variant and Invariant system II) Linear and Non Linear system	05	CO1	BT1
d	Write a Merits and Demerits of positive and negative feedback system.	05	CO1	BT2
Q.2	Attempt any two questions.			
a	To find the Transfer Function from given Block Diagram.	05	CO2	BT3
b	Define the following terms : 01) Transfer Function 02) Signal Flow Graph.	05	CO2	BT1
c	To calculate the Transfer Function from given Signal Flow Graph.	05	CO2	BT3
d	Explain the Mason's Gain Formula, with suitable example (Assume any example).	05	CO2	BT2



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Electrical Engineering
In-Semester Examination - I
Academic Year: 2025-26 (Semester – I)

Class: Third Year	Subject: Power System Engineering	Subject Code – PCC-EE301
Date: 18/08/2025	Time: 15:00 to 15:45	Max. Marks: 20

Note:

- i. All questions are compulsory
- ii. Assume suitable data whenever necessary and state it clearly.
- iii. Mobile phones and programmable calculators are strictly prohibited.
- iv. Exchange/Sharing of stationery, calculator etc. not allowed.

Q.N.	Questions	Marks	CO	BT
Q.1	a) Explain structure of a power system with SLD and concept of grid and microgrid?	6	CO-1	BT-1
	b) Write a short note on types of distribution system.	4	CO-1	BT-1
Q.2	a) Derive the expression for Flux linkages due to internal flux.	6	CO-2	BT-2
	b) Explain about skin and proximity effects?	4	CO-2	BT-2

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Describe the evolution, structure, and operation of modern power systems including conventional and renewable generation, transmission, and distribution systems; and analyse AC power flow and reactive power in three-phase circuits. (Understand, Analyze)
CO 2	Analyze and compute the inductance and capacitance of overhead transmission lines using concepts of GMD and GMR, and evaluate the effects of skin effect, proximity effect, and Ferranti effect. (Analyze, Evaluate)
CO 3	Model and evaluate electrical parameters of transmission lines using ABCD constants, and compute performance metrics such as voltage regulation, efficiency, and surge impedance loading. (Apply, Evaluate)
CO4	Explain and simulate the performance of transformers and synchronous machines, apply per-unit modelling, and interpret machine behaviour under normal and fault conditions. (Understand, Apply)
CO5	Analyze the mechanical aspects of transmission lines, including sag calculation, insulator string efficiency, and the impact of environmental conditions such as wind and ice on line performance. (Analyze, Evaluate)
CO6	Design and assess AC and DC distribution systems based on layout schemes, voltage drop, and system reliability; solve numerical problems related to performance evaluation. (Apply, Analyze)



Class: SY B.Tech.

Course: Electromagnetic Fields

Course Code: PCC-EE204

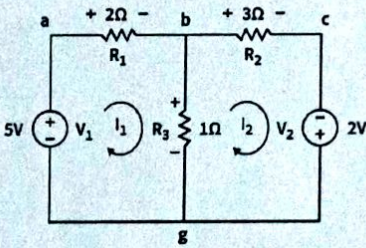
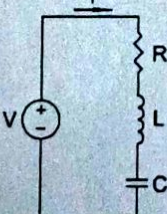
Date: 19.08.2025

Time: 11:00 A.M to 11:45 A.M

Max. Marks: 20

Note:

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Q.N.		Question Statement	Marks	CO	BT
Q.1	a	Define the basic circuit parameters: Current and Voltage, including their units.	2	CO1	BT1
	b	Given a 10 V voltage source in series with a $5\ \Omega$ resistor, use source transformation to find the equivalent current source and the value of the parallel resistor.	3	CO1	BT3
Q.2	a	Define the following terms: (i) Node, (ii) Branch, (iii) Graph, (iv) Tree and (v) link or chords	5	CO1	BT1
Q.3	a	For the circuit shown in the figure 1, there are two voltage sources $V_1 = 5V$ and $V_2 = 2V$. Currents I_1 and I_2 pass through loops 1 and 2 respectively with resistances $R_1 = 2\ \text{ohms}$, $R_2 = 3\ \text{ohms}$ and $R_3 = 1\ \text{ohms}$. Using KVL, find currents I_1 and I_2.  Figure. 1	6	CO2	BT3
	b	The given figure 2 represents the RLC circuit. Find the output voltage of the given circuit in s-domain. (Assuming the initial conditions are 0.)  Figure 2.	4	CO2	BT3

== ALL THE BEST ==